

ENCN445: Environmental Fluid Mechanics

Coastal Engineering Tutorials:

Modules One and Two

Problem One:

A fishing boat on the ocean surface finds itself moving up and down. They find that the vertical distance between the minimum and maximum elevation is 1 m and it takes 5 s for the boat to move from a trough to a crest. From looking at the movement of nearby buoys they estimate that the wavelength (from crest to crest) of the wave is about 200 m. From a depth sounder they know that the water depth is 600 m.

A *Calculate the wave amplitude, wavenumber and angular frequency of the wave.*

The wave amplitude is half the trough to crest distance so is

$$\eta_0 = 0.5 \times 1 = 0.5 \text{ m.}$$

The wavenumber is given by

$$k = (2\pi/\lambda) = (2\pi)/200 = 0.0314 \text{ rad/m.}$$

The angular frequency is given by

$$\omega = 2\pi/T = (2\pi)/(5 \times 2) = 0.628 \text{ rad/s.}$$

B *Can the small wave approximation be used on this wave?*

The small wave approximation requires both

$$\frac{\eta_0}{\lambda} \ll 1$$

and

$$\frac{\eta_0}{h} \ll 1.$$

For this wave

$$\frac{\eta_0}{\lambda} = \frac{0.5}{200} = 0.0025 \ll 1 \quad \text{and} \quad \frac{\eta_0}{h} = \frac{0.5}{600} = 0.0008 \ll 1$$

so the small wave approximation is valid.

C Calculate the phase velocity based on the measurements made on the boat and the theoretical wavespeed based on the dispersion relation.

Based on measurements from the boat, the phase velocity is given by

$$c = \frac{\lambda}{T} = \frac{200}{5 \times 2} = 20 \text{ m/s}.$$

We know that this is a deep water wave ($h/\lambda = 0.333$). Therefore we expect the phase velocity to be given by

$$c = \sqrt{\frac{g}{k}} = \sqrt{\frac{9.81}{0.0314}} = 17.7 \text{ m/s}$$

D Comment on the difference between the two phase velocities calculated in part C. Justify which phase velocity you think is more accurate and adjust the calculations of the basic wave properties in part A accordingly.

The two phase velocities disagree by around 10%. This suggests that one of the calculations is wrong. We could test the deep water assumption by evaluating $\tanh(kh)$. For a deep water wave we expect this term to tend towards 1. Evaluating $\tanh(kh)$ with $k = 0.0314 \text{ rad/m}$ and $h = 600 \text{ m}$ gives 1.0000000000000000. Clearly the deep water assumption is a very reasonable assumption!

We can use the dispersion relation to assess the accuracy of the measurements made on the boat. We know that

$$\omega^2 = gk \tanh(kh)$$

Assuming that our measurement of the wavelength is correct we can see what the period would have to be:

$$\omega = \sqrt{gk \tanh(kh)} = \sqrt{9.81 \times 0.0314} = 0.555 \text{ rad/s}$$

where we have let $\tanh(kh) = 1$. This is significantly different from what we calculated in part A based on the measurements from the boat. Thus, the measurements from the boat imply a wave that **can't exist** (unless gravity was significantly stronger).

Although there are going to be errors with the measurement of both the wave period and wavelength from the boat, the wavelength seems to be less reliable. It is much easier to time how quickly you are going up and down than to estimate the length of a wave over hundreds of metres. Therefore we will assume that $\omega = 0.628 \text{ rad/s}$. We can calculate the wavenumber as

$$k = \frac{\omega^2}{g} = \frac{0.628^2}{9.81} = 0.0402 \text{ rad/m}$$

where we have again assumed that $\tanh(kh) = 1$. Finally we calculate the wavelength as

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.0402} = 156 \text{ m}$$

Finally the phase velocity can be calculated as

$$c = \frac{\omega}{k} = \frac{0.628}{0.0402} = 15.6 \text{ m/s}$$

which is significantly lower than either of the estimates given in C.

Problem Two:

A long wave signal is observed 200 km offshore from a populated beach. Where the wave was observed the ocean depth is 2500 m. The measured wavelength is 275 km and the wave period is 1757 s. The amplitude of the wave is only 10 cm.

A *Without making any calculations do you think the people on the beach should be concerned by this relatively small amplitude wave that's still 200 km offshore?*

The people on the beach should definitely be worried. This wave has the characteristics of a tsunami (very long period and wavelength). This means that it will travel very quickly and reach the beach soon. Although the amplitude is currently small, the amplitude of waves tends to increase as they approach the shore so this could be a significant tsunami.

B *Classify this wave as a shallow or a deep water wave based on the measured characteristics. Use this to calculate the group velocity (i.e. the velocity with which the wave energy will approach the shore) and how long it will take for the wave energy to reach the shore at its current velocity.*

We can calculate the ratio of the fluid depth to the wavelength to determine if the wave is a deep or shallow water wave.

$$\frac{h}{\lambda} = \frac{2500}{275000} = 0.00909.$$

We usually classify waves as shallow water waves if this ratio is smaller than 0.4 so this wave is definitely a shallow water wave.

The group velocity of a shallow water wave is the same as its phase velocity and is given by

$$c = \sqrt{gh} = \sqrt{9.81 \times 2500} = 157 \text{ m/s}$$

Therefore the time it will take for the wave to reach the ocean is

$$t = \frac{d}{c} = \frac{200000}{157} = 1274 \text{ s} \approx 21 \text{ minutes.}$$

C *Calculate the energy density of the wave. Compare this to a typical swell wave such as the one examined in the previous problem ($\eta_0 = 0.5 \text{ m}$; $k = 0.0402 \text{ rad/m}$; $\omega = 0.628 \text{ rad/s}$)*

We can use the simple formula for the energy density:

$$E = \frac{\text{PE} + \text{KE}}{\lambda} = \frac{1}{2} \rho g \eta_0^2.$$

For our measured wave this gives

$$E = \frac{1}{2} \rho g \eta_0^2 = \frac{1000 \times 9.81 \times 0.1^2}{2} = 49.05 \text{ J/m}^2$$

The units of J/m^2 are because this is an energy density per unit length of the wave and we are also considering a two-dimensional wave so the dimension ‘into the page’ also needs to be accounted for. For a typical swell wave we get

$$E = \frac{1}{2}\rho g \eta_0^2 = \frac{1000 \times 9.81 \times 0.5^2}{2} = 1226.25 \text{ J/m}^2$$

Thus a swell wave has an energy density that is 25 times as high as our tsunami.

D *In practice the wave with ‘shoal’ as it approaches the shore causing its wavelength to decrease. To help compensate for this without actually calculating the shoaling directly, calculate the energy flux and the energy carried by a single wavelength for both the measured wave and our typical swell wave. You can still treat the wave as two-dimensional.*

The energy flux of our measured wave is given by

$$\text{Flux} = c_g E = 157 \times 49.05 = 7701 \text{ J/m/s}$$

Our swell wave has a wavelength given by

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.0402} = 156 \text{ m}$$

Since $156 \ll 2500$ we can treat the wave as a deep water wave. For deep water waves the group velocity is given by

$$c_g = \frac{1}{2}\sqrt{\frac{g}{k}} = \frac{1}{2}\sqrt{\frac{9.81}{0.0402}} = 7.8 \text{ m/s}$$

Thus our energy flux is

$$\text{Flux} = c_g E = 7.8 \times 1226.25 = 9565 \text{ J/m/s}$$

This shows why tsunami aren’t particularly significant away from shore. They typically have very small amplitudes which means that the energy flux is comparable with a single swell wave.

We will now look at the total energy carried in a single wavelength. It is just the energy density multiplied by the wavelength. Thus, for our measured wave we have

$$E\lambda = 49.05 \times 275000 = 13,500,000 \text{ J/m} = 13,500 \text{ kJ/m}$$

and for a typical swell wave we have

$$E\lambda = 1226.25 \times 156 = 192,000 \text{ J/m} = 192 \text{ kJ/m}.$$

This helps to demonstrate that the only reason that tsunami are so destructive is because the wave shoals as it approaches the coast. The wavelength decreases and the wave amplitude increases which results in a huge amount of energy being released in a short period of time.

Problem Three: *Describe the key assumptions that were used to derive our wave equations and which ones are the most restrictive.*

We assume that the **fluid is of constant depth**. For most waves this isn't an issue at all as the wave is 'unaware' of the fluid depth. Physically this means that the wave velocities are zero at some distance much above the bottom of the fluid. These waves are called deep water waves and are completely insensitive to the depth of the fluid. For shallow water waves our solutions are locally valid but we still need to be careful with this assumption. Provided the changes in fluid depth are small relative to the wavelength, our locally valid equations can be useful but if there were steep gradients in the sea floor we would need to do things differently.

We have assumed that the waves are **inviscid and irrotational**. These assumptions allow us to formulate our equations so are very important to the results we obtain. Luckily the solutions are typically good. Similarly to our constant fluid depth assumption we need to be careful close to the shore. Here waves start breaking and turbulent processes become important. Generally this assumption is fine though.

We assume that the waves are **two-dimensional**. This is a significant assumption. Often waves will be formed at a point and the spread radially. The radial spreading will change how much energy is carried in the wave per unit length 'into the page'. Once again the solution will remain correct over small spatial scales however so we can use it locally. Here, locally depends on the distance from the source so our solutions become better closer to shore where we care about them the most.

We have assumed that the waves are **small amplitude**. By this we mean that the wavelength and the fluid depth are both significantly larger than the wave amplitude. This is almost always true in the open ocean and is clearly not true as waves approach the shore (where the fluid depth goes to zero).

Generally our assumptions tend to become worse as we get closer to shore. This is not unsurprising given our simple initial geometry. Anyone who's been to the beach on a stormy day would probably believe that breaking waves are a complicated process to model. Nonetheless, our model of open ocean waves provides a lot of insight and can describe to a high degree of accuracy many important processes such as the dispersion of waves in frequency bands, the propagation of a tsunami across the ocean, and the amount of energy carried by a given wave field.